INCH-POUND

MIL-C-46409D (AR)
10 April 1991
SUPERSEDING
MIL-C-46409C (MU)
14 June 1976

MILITARY SPECIFICATION

CARTRIDGE, CALIBER .38, SPECIAL, BALL, M41

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts, and the assembly, and packing of Cartridge, .38, Special, Ball, M41.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550	- Ammunition, General Specification For
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition for
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1305

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STANDARDS

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-636	- Visual Inspection Standards and Inspection Procedures for Small Arms Ammunition through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing, and Marking of Small Arms Ammunition
MIL-STD-1168	- Lot Numbering of Ammunition

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 191 11-5094).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.5)

US ARMY ARMAMENT RESEARCH DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

7553580	- Cartridge, Caliber .38 Special, Ball, M41
7643674	- Classification of Cartridge Case Defects
9396206	- Packing and Marking for Wire bound Box Assembly for Cartridges, Small Caliber
LI-7553580	- Inspection Equipment List for Cartridge, Caliber .38, Special, Ball M41

PUBLICATIONS

US Army Munitions Command
TECP 700-700 Vol. III - Manual of Test Methods for Small Arms Ammunition

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Cartridges. The cartridges shall comply with all requirements specified on drawing (Dwg.) 7553580, all associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Bullet extraction. The force required to extract the bullet from the cartridge case shall be not less than 100 pounds.

3.4 Residual stress. The cartridge case shall not split when subjected one percent mercurous nitrate solution for 15 minutes.

3.5. Velocity. The average velocity of the sample cartridges, conditioned at 68° to 72° Fahrenheit (F) (ambient) shall be 950 feet per second (ft/sec) plus or minus 45ft/sec at 15 feet from the muzzle of the weapon. The standard deviation of the velocities shall not exceed 27 ft/sec.

3.6. Chamber pressure. The chamber pressure of the cartridges, temperature conditioned and fired at the temperature prescribed, shall not exceed the following:

Temperature Degree (°) Fahrenheit (F)	Average Pressure Pounds per square inch (PSI) Maximum	Single Shot pressure (PSI) Maximum
-40° to -45°	20,000	24,000
68° to 72°(ambient)	17,000	20,000
125° to 130°	20,000	24,000

3.7. Accuracy. The average of the diagonals (D) of all targets of the sample cartridges fired at 50 yards shall not exceed 7.46 inches.

3.8. Fuction and casualty. The cartridge shall function without casualty at ambient temperatures, at minus 40 degrees to minus 45 degrees F and at plus 125 degrees to plus 130 degrees F.

3.9. Waterproof. The cartridge shall not release more than one air bubble when subjected to a pressure differential of 5 psi for 15 seconds.

3.10 Cartridge elongation. The cartridge chambered but not fired, shall not, after the firing of other cartridges chambered in the same cylinder exceed 1.565 inches in length.

3.11 Primer sensitivity. The energy imparted by a steel ball 1.94 ounces falling 11 inches shall cause initiation of the primer. The energy imparted by a steel ball 1.94 ounces falling 1 inch shall not cause initiation of the primer.

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3.12 Propellant weight. Each cartridge shall contain a propellant charge weight of 3 grains minimum when loaded with HPC type propellant or 3.5 grains minimum when loaded with SR type propellant.

3.13 Data requirements. The contractor shall generate and deliver data in accordance with DD Form 1423 in the contract. This shall include submission of the detailed inspection plan for the item for review by the Government Technical Agency.

3.14 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.15 Workmanship. The requirements for workmanship shall be in accordance with MIL-A-2550 and as specified on the applicable drawing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

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4.3 First article inspection.

4.3.1 First article sample. The First Article Sample shall be subjected to the first article inspection to determine contract compliance. The first article sample shall be representative of the manufacturing methods and processes to be used for quantity production. The first article samples shall be submitted and tested prior to the beginning of production.

4.3.2 Submission The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of Table I. The first article sample shall consist of the following items in sample quantities as indicated

PART DESCRIPTION	DRAWING	QUANTITY
Cartridges	C7553580	2798
Empty Primed Cartridge cases (no propellant or projectile)	(as appropriate)	550

4.3.3 Inspection to be performed. First article assemblies and components may be subjected by the Government to any or all of the examinations and tests specified in Table I and to any or all requirements of the applicable drawings. All cartridges shall have passed the criteria specified in 4.4.2.4 prior to being tested in Table I.

4.3.4 Rejection. The First Article quantity shall be rejected upon failure of any assembly or component in the sample to comply with any visual or dimensional characteristic or upon failure to comply with the first sample accept-reject criteria for any test. No second sample tests shall be permitted on the First Article sample. The Government reserves the right to terminate its inspection whenever a component or assembly fail to comply with any visual or dimensional characteristic or components or assemblies fail to comply with the criteria for any test. MIL-A-48078 shall apply.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46409D (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Cartridge, Caliber .38, Special, Ball M41				C7553580
					NEXT HIGHER ASSEMBLY
					C7553580
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for defects	Min 2798		3.2	4.4.2.4
	Primer Sensitivity 1/	550		3.11	4.4.3.1
	Bullet extraction 2/	20		3.3	4.5.1
	Residual stress (mercurous nitrate) 5/	20		3.4	4.5.2
	Waterproofness 3/	20		3.9	4.5.8
	Cartridge elongation 2/	10		3.10	4.5.9
	Propellant weight 4/	100		3.12	4.5.10

NOTES:

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46409D (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Cartridge, Caliber .38, Special, Ball M41				C7553580
					NEXT HIGHER ASSEMBLY
					C7553580
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Ballistic Tests				
	Velocity 5/	20		3.5	4.5.3
	Chamber Pressure 5/			3.6	4.5.4
	68°F to 72°F	20			
	-40°F to -45°F	20			
	125°F to 130°F	20			
	Accuracy 5/	100		3.7	4.5.5
	Function and Casualty 6/			3.8	4.5.6
	68°F to 72°F	816			
	-40°F to -45°F	816			
	125°F to 130°F	816			

NOTES:

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NOTES

1/ Empty primed cartridges cases (no propellant or projectile). If primer sensitivity test fails to meet the requirement on the first sample the fist article shall be rejected. No second sample shall be allowed.

2/Failure of any cartridge to comply with the applicable requirement shall be cause for rejection of the first article

3/ Failure of four or more cartridges to comply with the applicable requirement shall be cause for rejection of the first article.

4/ If one or more cartridges contain less than the minimum required propellant weight, the first article shall be rejected.

5/Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the first article.

6/ The first article shall be rejected when function and casualty defects, plus firing defects exceed the acceptance number for the frost sample in Table II.

4.4 Quality conformance inspection.

4.4.1 Lot.

4.4.1.1 Submission of product. The product shall be submitted in accordance with MIL-STD-105.

4.4.1.2 Lot size. The size of an ammunition lot shall be not less than 200,000 nor more than 500,000 cartridges.

4.4.1.3 Lot identification. Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number in accordance with MIL-STD- 1168. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.4.2 Examination. Each charged cartridge case shall be examined for propellant charge, (wt) minimum, requirement of 3.12. Method of examination shall be approved by the contracting officer. The equipment design shall provide control(s) that will assure rejection of defective propellant charge (wt), in the cartridge case. On hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, 4.4.2.4 using applicable sampling plans and acceptance criteria of MIL-STD-105. The acceptable quality level (AQL) for the major class shall be 0.25 percent and the AQL for the minor class shall be 1.50 percent.

4.4.2.1 Classification of defects. The classification of defects shall be as specified in 4.4.24.

4.4.2.2 Order of test. All cartridges shall have been submitted to and passed the criteria specified in 4.4.2.4 prior to being tested n 4.4.3.

4.4.2.3. Rejection. All non-conforming cartridges shall be classed defective and rejected. Non-conforming cartridge lots shall be rejected.

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QUALITY CONFO. ANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46409D (AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
4.4.2.4	Cartridge, Caliber .38, Special, Ball M41				C7553580
					NEXT HIGHER ASSEMBLY
					C7553580
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Case split in K, L, or M location 1/ 2/		100%	3.2	Visual
2.	Perforated case 1/		100%	3.2	Visual
3.	Primer missing 1/		100%	3.2	Visual
4.	Primer cocked 1/		100%	3.2	Visual
5.	Primer inverted 1/		100%	3.2	Visual
Major					
101.	Cartridge corroded, or stained 1/		.25%	3.2	Visual
102.	Case split in I, S or J location 1/ 2/		.25%	3.2	Visual
NOTES:					

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AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
4.4.2.4	Cartridge, Caliber .38, Special, Ball, M41		C7553580
			NEXT HIGHER ASSEMBLY C7553580

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
103.	Bullet jacket split 1/		.25%	3.2	Visual
104.	Bullet loose 1/		.25%	3.2	Visual
105.	Bullet, defective cannelure 1/		.25%	3.2	Visual
106.	Primer loose 1/		.25%	3.2	Visual
107.	Total length, cartridge, max.		.25%	3.2	Gage
108.	Cartridge profile failure (requiring more than 5 pounds dead weight to insert in profile and alignment gage)		.25%	3.2	Gage
109.	Case, thickness of head, max.		.25%	3.2	Gage
110	Depth of primer, min.		.25%	3.2	Gage
111.	Case not crimped in cannelure 1/		.25%	3.2	Visual
112.	Case dent 1/		.25%	3.2	Visual
113.	Case, draw scratch 1/		.25%	3.2	Visual
114.	Bullet, scaly metal 1/		.25%	3.2	Visual
115.	Scaly metal (case) 1/		.25%	3.2	Visual
116.	Case, head, rounded 1/		.25%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.4.2.4	Cartridge, Caliber .38, Special, Ball, M41				C7553580
					NEXT HIGHER ASSEMBLY
					C7553580
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201.	Cartridge, discolored, dirty, oily smeared 1/		1.5%	3.2	Visual
202.	Case, draw scratch 1/		1.5%	3.2	Visual
203.	Case, scratch 1/		1.5%	3.2	Visual
204.	Case, scaly metal 1/		1.5%	3.2	Visual
205.	Chamfer on head (rim) missing 1/		1.5%	3.2	Visual
206.	Case, wrinkle 1/		1.5%	3.2	Visual
207.	Illegible or missing head stamp 1/		1.5%	3.2	Visual
208.	Defective head 1/		1.5%	3.2	Visual
209.	Defective mouth 1/		1.5%	3.2	Visual
210.	Bullet dent 1/		1.5%	3.2	Visual
211.	Bullet scratch 1/		1.5%	3.2	Visual
212.	Nicked or dented primer 1/		1.5%	3.2	Visual
213.	No waterproofing (primer pocket joint) 1/		1.5%	3.2	Visual
214.	Cartridge, total length, min.		1.5%	3.2	Gage
215.	Case, diameter of head		1.5%	3.2	Gage
216.	Case, thickness of head, min.		1.5%	3.2	Gage
217.	Depth of primer, max.		1.5%	3.2	Gage
218.	Workmanship		2.5%	3.15	Visual

NOTES:
1/ Refer to .45 caliber cartridge section of MIL-STD-636 for visual standards of defects.
2/ Refer to Drawing C7643674 for location of defect.

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4.4.2.5 Packaging, packing and marking inspection. During or immediately prior to the packaging operation, 100% examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a high pressure test, dummy or blank cartridge shall be classed as a critical defect. Occurrence of any type other than those listed shall be classed as a major defect. All nonconforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

4.4.3 Tests. The tests listed in Table I and the 4.4.3.3 shall be conducted in accordance with methods and procedures specified in 4.5.

4.4.3.1 Primer sensitivity test. Prior to performance of the tests cited in 4.4.3.3 a sample of primed cases shall be tested for primer sensitivity. The sample shall be selected from the entire lot or from individual sub-lots, at the option of the contractor, provided that the size of each sub-let is not less than 200,000. The test sample shall be representative of the entire lot or sub-lot as applicable, and shall consist of the quantity required for performance of a complete run-down test wherein fifty primers are tested at each height. If the average critical height (H) plus five standard deviations (5 sigma) exceeds 11 inches, or if the average critical height (H) minus two standard deviations (2 sigma) is less than 1 inch, the lot or sub-lot shall be rejected subject to performance of a second complete run-down test wherein one hundred primers are tested at each height. Failure of the second sample to comply with the above criteria shall be cause for rejection of the lot or sub-lot as applicable.

4.4.3.2 Test samples. The quantities for the various tests shall be as specified in Table I and 4.4.3.3. Only cartridges having met the visual and dimensional requirements shall be used in the ballistic tests and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges selected shall be thoroughly mixed before being divided into samples for the various tests.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46409D (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.3.3	Cartridge, Caliber .38, Special, Ball, M41		C7553580
			NEXT HIGHER ASSEMBLY: C7553580

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
		Min			
	Examination for defects	2798		3.2	
	Primer sensitivity 1/	550		3.11	4.4.3.1
	Bullet extraction 2/	20		3.3	4.5.1
	Residual stress (mercurous nitrate) 5/	20		3.4	4.5.2
	Waterproofness 3/	20		3.9	4.5.8
	Cartridge elongation 2/	10		3.10	4.5.9
	Propellant weight 4/	100		3.12	4.5.10
	Ballistic tests				
	Velocity 5/ 7/	20		3.5	4.5.3
	Chamber pressure 5/			3.6	4.5.4
	68°F to 72°F	20			
	-40°F to -45°F	20			
	125°F to 130°F	20			
	Accuracy 5/	100		3.7	4.5.5
	Function and casualty 6/			3.8	4.5.6
	68°F to 72°F	816			
	-40°F to -45°F	816			
	125°F to 150°F	816			

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NOTES

1/ Empty primed cartridge cases (no propellant or projectile).

2/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample may be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

3/ Failure of nine or more cartridges to comply with the applicable requirement shall be cause fur rejection of the lot. If more than four, but less than nine cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample may be tested. The lot shall be rejected if, in the combined first and second sample, nine or more cartridges fail to comply with the applicable requirement.

4/ If one or more cartridges contain less than the minimum required propellant weight, the lot shall be rejected.

5/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges in the first sample. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

6/ The lot shall b rejected when function and casualty defects, plus firing defects exceed the acceptance number for the cumulative sample in Table II. If the number of defects found in the first test exceeds the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample, consisting of double the quantities specified under function and casualty test, shall be fired in all the service weapons specified therefor. This procedure shall apply regardless of the weapon or weapons in which the firing defects occurred in the first test. If the total number of defects in the combined first and second sample exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

7/ Test will be conducted simultaneously with ambient chamber pressure test.

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Table II

Function and Casualty Defect Classification Accept-Reject Criteria

Defect	First Sample	Acceptance Number Cumulative (First & Second Samples)
1. Misfire 6/		
a. No vent hole, or obstruction in vent area	0	-
b. Other	1	2
2. Bullet remaining in bore	0	-
3. Primer leak		
a. Perforation in firing pin indent in primer cup	19	45
b. Escape of gas through primer cup other than a.	13	29
c. Escape of gas around primer cup 2/	50	165
d. Loose primer 3/	19	45
e. Blown primer 4/	0	1
4. case casualty		
a. Longitudinal split 5/		
(1) Mouth (I)	38	95
(2) Body (J)	28	69
(3) Body (K)	1	2
(4) To head (L)	0	1
(5) Through head (M)	0	1
b. Circumferential rupture 5/		
(1) Mouth (I)	7	16
(2) Partial (J)	0	1
(3) partial body (K)	0	1
(4) partial head (L)	0	1
(5) Complete	0	1
5. Weapon stoppage 7/	0	1

1/ No retest permitted. Lot shall be rejected.

2/ Gas escape around more than fifty percent of periphery of cup.

3/ Independent movement of primer in case packet or primer falls out of packet.

4/ Primer not in packet after firing and both head of cartridge case and packet enlarged and distorted (determined visually).

5/ For location of defects indicated by letters in parentheses, see Drawing C7643674.

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6/ Each cartridge that misfires shall be disassembled and examined for presence a vent hole in the primer pocket, or any obstruction in the vent hole are of the primer pocket that can be assignable as the cause for misfire. If the vent hole is missing or obstructed, the lot shall be rejected with no second sample permitted

7/ Weapon stoppage, with the exception of misfire, observed in all test shall be included.

4.4.4 Inspection equipment. The examination and tests shall be made using equipment listed in LI-7553580. Unless otherwise specified, acquisition, maintenance and disposition of inspection equipment shall be in accordance with MIL-I-45607.

4.5 Methods of inspection.

4.5.1 Bullet extraction. The test shall be conducted in accordance with Appendix A". the rate of travel of the test head, in the bullet extractor machine, shall be not less that three (3) inches nor more than six (6) inches per minute.

4.5.2 Residual stress (mercurous nitrate). The test shall be conducted in accordance with Appendix "B".

4.5.3 Velocity. The test shall be conducted in accordance with Appendix "C".

4.5.4 Chamber pressure. the tests shall be conducted in accordance with Appendix "D" or Government approved equivalent. The test shall be conducted at -40° to -45°F ambient and 125° to 130°F. An equal number of cartridges shall be tested at each of the three temperatures. The obturator shall be of the wax-filled type (see 6.6).

4.5.5 Accuracy. The test shall be conducted in accordance with Appendix "E".

4.5.6 Function and casualty. the test shall be conducted in accordance with Appendix "F". The cartridges shall be fired single action and double action at the temperatures indicated below:

Temperature (F)	-40° to -45°	Ambient	125° to 130°
Single Action	408	408	408
Double Action	408	408	408

The test quantities shall be distributed equally in so far as possible among the prescribed test weapons. The Government Contracting Officer shall designate the weapons in which the Function and Casualty test shall be performed depended upon New Weapon Availability and Weapon Inventory Supplies. The weapons shall be selected from the following:

a. Colt, Revolver, Caliber .38 Special, 4 Inch Barrel

b. Smith& Weston, Revolver, Caliber .38 Special 4 Inch Barrel

c. Ruger, Revolver, Caliber .38 Special, Security Six, 4 Inch Barrel

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4.5.7 Temperature (High and Low). The cartridges for the high and low temperature test shall be conditioned in accordance with Appendix "G".

4.5.8 Waterproof. the test shall be conducted in accordance with Appendix "H".

4.5.9 Cartridge Elongation. The test shall be conducted in accordance with Appendix "I". The cartridges shall be scrapped after the test.

4.5.10 Propellant weight. The bullet of the test sample shall be removed from the cartridges, then the propellant from each case shall be weighed. Weights shall be taken on a balance scale that is calibrated to the nearest 0.01 of a grain. results of each propellant weight shall be recorded for Government verification.

4.5.11 Primer sensitivity. the primer sensitivity test shall be conducted on empty primed cases in accordance with TECP 700-700 Vol III. The primed cases may be obtained by either of the following methods provided that the method is consistent throughout entire sample:

a. Preferred method. primed cases shall be obtained by selecting samples from primed, unloaded cartridges (prior to loading) providing that test samples are representative of the loaded cartridge lot.

b. Alternate method. primed cases may be obtained for special cases, by disassembly of complete cartridges which have been randomly selected from the loaded cartridge lot. This disassembly shall be accomplished in such a manner as to cause the least possible distortions of the cartridge case.

5. PACKAGING

5.1 Packing - Level A. (worldwide shipment) - The cartridge shall be packed in accordance with Drawing 9396206 or as required by contract.

5.2 Marking and labeling. Packing boxes shall be marked and labeled in accordance with the applicable drawing cited in 5.1. Unless otherwise specified, marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The cartridges covered by this specification are intended for use in modern .38 caliber weapons.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

c. Type and level of packing of this specification

d. Provision for submission of Inspection Equipment Designs (DI-R-1714).

e. Provision for submission of detailed inspection plan for the cartridge (DI-R-4803) (See 6.8).

f. Provisions for submission of acceptance inspection results for each lot of ammunition presented to the Government (See 6.8).

6.3 Submission inspection equipment designs for approval. Submit inspection equipment designs as required by the contract to Commander, ARDEC, ATTN: AMSMC-QAF-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified in the contract Data Requirements List, DD Form 1423 in the contract.

6.4 Hazard notice. The cartridge described herein and some of its components are flammable and/or explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measure to prevent fire, explosion, adverse environment, rough handling, corrosive atmosphere, electrically induced incidents. Such measures shall include the employment of an effective safety program that addresses the inherent hazards associated with the cartridge.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, US Army Armament Research and Development Center (ARDC) drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.6 Obturating cups (wax filled). Prior to use, each obturating cup shall be filled with wax. The proof technician shall fill the cup in such manner that excess wax extends over the lop of the cup, the cup is then placed on a smooth surface mouth and down and rubbed back and forth until the wax is flush with the lips of the cup. Overfilled or underfill cups shall not be used., Obturating cups (wax filled) shall not be lubricated prior to use. Wax ingredients and method of preparation are as follows:

6.6.1 Wax ingredients.

Beeswax	234.0 Grams
Paraffin	6.0 Grams
Vaseline	16.0 Grams
Castor Oil	14.4 Grams
Lead Oxide (Red Pb_3O_4)	72.0 Grams
Iron Oxide (Fe_3O_4)	24.0 Grams
Rosin	5% by Volume

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6.6.2 Preparation. The ingredients are weighed out in a casserole and heated on a steam bath until the waxes are melted. The mixture is then removed and stirred vigorously until slightly warm. The Wax is then rolled out on a flat surface into sticks.

6.7 Process deviation. A process deviation is defied as a change in the initial approved basic method of manufacture, This also includes any operational changes i.e., change in sub-contractor (s), changes which may alter the metallurgical or physical properties of the item which were approved initially in the basic method of manufacture. any manufacturing deviation, after the "First Article sample" has been approved should be submitted to the technical agency for approval.

6.8 Submission of test data. Submit test data in accordance with the contract.

6.9 Submission of detailed inspection plan for review. The detailed inspection plan, and any changes thereto, for the cartridge shall be submitted to the Government as required by the contract.

6.10 Subject term (key word) listing.

Ammunition
Cartridge, Caliber .38, Special, Ball
Military Specification
M41

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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APPENDIX "A"

BULLET EXTRACTION TEST PROCEDURE

10.0 PURPOSE

The Bullet Extraction Test determines the force required to pull a bullet from its cartridge case. It is used as a measure of the uniformity and efficiency with which bullets are secured in the cases.

20.0 EQUIPMENT

Equipment listed in the Bullet Extraction section of the appropriate Inspection Equipment list shall be used.

30.0 TEST PROCEDURE

30.1 Preparation for test

30.1.1 The testing machine should be calibrated either weekly or prior to each occasion of use, whichever is less frequent. Calibration points shall included at least 30 pounds, 60 pounds, 100 pounds, 150 pounds and 250 pounds. If calibration errors exceed three pounds at scale readings below 200 pounds or six pounds at scale readings above 200 pounds, then the necessaries corrective action shall be taken to reduce the errors below these limits before the bullet extraction test is conducted. The method of calibration shall be calibrated proof rings or calibrated spring balances which shall yield values within ± 1% of true values.

30.1.2 Each test cartridge shall be secured in the jig fixture and, using a standard drill press with a 1/8" drill, a hole shall be drilled completely through the bullet.

30.1.3 The number of cartridges as prescribed by the applicable specification are placed in a recessed holding block which shall be placed at a point convenient to the technician.

30.2 Conducting the test

302.1 The cartridge shall be inserted into the case holding block on the pulling head and aligned i the jaws. The jaws shall then se secured to the bullet just above the mouth of the case.

30.2.2 The load shall be applied.

30.2.3 When the bullet has been extracted from the case, the machine shall be stopped. The tensile indicator records the force required to extract the bullet.

30.2.4 The case and bullet shall then be removed form the case-holding block and jaws.

30.2.5 The case with the propellant should be placed upright in a recessed holding block and the extracted bullet placed point down in the mouth of the case. If the components not needed, they shall be scrapped in accordance with local safety requirements.

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40.0 RECORDING OF DATA

40.1 Test sheets shall show the following:

a. Force required to extract each bullet.

b. Grand average of the individual values.

c. Minimum value.

d. Maximum value.

e. Testing machine data.

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APPENDIX "B"

RESIDUAL STRESS (MERCUROUS NITRATE) TEST PROCEDURE

10.0 PURPOSE

Mercurous Nitrate test is visual means of determining if stresses exist in brass cartridge cases that may cause the cases to crack or split under conditions of storage or during service usage.

20.0 EQUIPMENT

Equipment listed in Mercurous Nitrate section of the appropriate Inspection Equipment List shall be used.

30.0 MANDATORY SAFETY REQUIREMENTS

30.1 Food shall not be stored or eaten in vicinity in which these tests are conducted.

30.2 Acid resistant apron and gloves or the equivalent shall be worn by each test technician.

30.3 Face shields shall be worn at all times during the pouring or mixing of acids and water.

30.4 Asbestos or heat insulating gloves shall be worn during the heat volatilization phase of the test to facilitate handling.

30.5 During the entire period of volatilization, the oven door shall be closed.

30.6 Extreme care shall be exercised in the mixing of acid with water; this shall be accomplished by pouring the acid into the water.

30.7 Test shall be conducted under a canopy or hood having a forced draft ventilation system to remove the noxious fumes. The disposition of tested components shall be governed by local regulations.

40.0 TEST PROCEDURE

40.1 Preparation of Solution

40.1.1 Nitric-Acid Solution: Four hundred (400) milliliters of nitric acid (of specific gravity 1.42) are dissolved in five hundred (500) milliliters of distilled water at room temperature. To this solution, distilled water is added to bring the volume of the resulting solution to one liter. (The resultant specified gravity will be 1-25).

40.1.2 Mercurous Nitrate Solution: Ten (10) grams of mercurous nitrate and ten (10) milliliters of nitric acid (of specified gravity1.42) are dissolved in four hundred (400) milliliters of distilled water at room temperature. To this solution, distilled water is added to bring the volume of the resulting solution to one liter.

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40.2 Test of cartridges.

40.2.1 The cartridges are positioned in both nitric acid and mercurous nitrate solutions in a vertical position, with the head of the case down. The depth of the solution is adjusted until it completely covers the mouth of the case but does not cover more than one-half the body of the bullet.

40.2.2 One-half of the number of cartridges prescribed in the applicable specification for the Mercurous Nitrate test is submerged in the nitric acid solution. After thirty (30) seconds the cartridges are withdrawn, rinsed in water and the excess water removed. The cartridges are then submerged for fifteen (15) minutes in the mercurous nitrate solution. Immediately upon removal the surface of the cartridges case in examined under a magnification (10 to 15 diameters) for splits and cracks. All splits and cracks shall be reported.

40.3 Test of cartridge cases

40.3.1 The cases for this test are obtained by disassembly of cartridges. The primers must be fired prior to using the cases.

40.3.2 The cases are positioned in both nitric acid and mercurous nitrate solutions in a vertical position, with the head of the case down. The depth of each solution is adjusted until it completely covers the mouth of the case but does not rise more than one-half inch above the mouth of the cases.

40.3.3 Cases from the remaining half of the number of cartridges prescribed in the applicable specification for the Mercurous Nitrate Test (after the bullet and propellant have been removed and primer fired) are submerged in nitric acid solution.

After thirty (30) seconds, the cases are withdrawn, rinsed in water and then excess water removed. The cases are then submerged for fifteen (15) minutes in the mercurous nitrate solution. Immediately upon removal, the mercury on the surface of the case is volatilized by the application of heat and the surface of the case examined under magnification (10 to 15 diameters) for splits and cracks. All splits and cracks shall be reported.

NOTE:

(1) A split is defined as a separation of the metal entirely through the wall of the case

(2) A crack is a surface condition and represents a separation of the metal partially through the wall of the case. Cracks are not considered to be splits.

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40.3.4 A suggested method for determining splits and cracks follows:

Splits in the cartridge case, after the mercurous nitrate test are detected by filling the case with water(water temperature 70° to 100°F) to the mouth until convex meniscus condition exists and placing the thumb of the hand over the mouth of the case. If the case is split, this exerts sufficient pressure to force the water through the opening.

Splits in the assembled cartridge are not so easily detected unless the split is at the mount of the case and the bullet metal is seen through the split. If the split is in the body, the only way to determine whether it is split or crack is to disassemble and follow the procedure used for the case check.

50.0 RECORDING THE DATA

50.1 Results of the test shall be recorded.

50.1.1 All splits and cracks shall be reported.

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APPENDIX "C"

VELOCITY TEST PROCEDURE

10.0 PURPOSE

The Velocity Test is a precisely controlled test to determine the velocity uniformity and level of the ammunition lot under test.

20.0 EQUIPMENT

20.1 Equipment listed in the Velocity section of the appropriate Inspection Equipment List shall be used.

20.2 The firing range shall be arranged as shown on Figure C-1

20.3 Pier of mount of solid construction on which the test fixture assembly shall be mounted

30.0 USE OF STANDARD (REFERENCE) CARTRIDGES

30.1 Standard (reference) cartridges shall be used to establish range and equipment corrections prior to firing an ammunition lot for acceptance. One standard (reference) cartridge shall be fired for each cartridge of the test sample to be fired, up to twenty (20) cartridges. When the test consists of more than twenty (20) but not more than forty (40) cartridges, then twenty (20) standard (reference) cartridges shall be fired. When the test consists of more than forty (40) cartridges, then one standard (reference) cartridge shall be fired for every two (2) test cartridges.

30.2 After the required number of standards (reference) cartridges have been fired, the actual average velocity if the standard (reference) cartridges shall be compared with the assessed value. If the assessed value is higher than the actual average velocity of the standards (reference) cartridges, the difference is a plus correction and shall be added to the average velocity of the test cartridges. If the assessed value is lower than the actual average velocity of the standard (reference) cartridges, the difference is a minus correction and shall be subtracted from the average velocity if the test cartridges. If the assessed value and the actual average velocity of the standard (reference) cartridges are identical, then no correction is applied.

30.3 The velocity-barrel assembly will be acceptable provided the results obtained with the standard (reference) cartridges are within ± 35 feet-per-second of the assessed value.

30.4 It is suggested that control charts shall be maintained for record purposes, of the results from each barrel in which standards (reference) cartridges are fired. Every corrrection obtained shall be compared with the previous one to determine whether the test conditions remain stable to a certain degree.

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APPENDIX "C" (CONT'D)

40.0 TEST PROCEDURE

40.1 Pre-firing (Preparation for test)

40.1.1 The required number of test cartridges and standard (reference) cartridges shall be placed in a vertical position, primer-end down, in recessed holding blocks. The cartridges shall be permitted to come to a temperature of 60° to 80° F prior to being placed in the temperature-controlled room or contained The recessed holding blinks containing two series of cartridges shall be placed in a temperature-controlled room or container in such a manner that all cartridges arc subjected to a uniform temperate for a minimum of two hours, prior to firing. The container or room shall be maintained at a temperature of 70°F, ± 2° F, with a relative humidity of 60 percent, ± 5 percent and be of sufficient capacity to allow free circulation of air.

40.1.2 The velocity-barrel assembly is assembled in the test fixture, on the mount. The chamber and bore of the barrel are wiped dry. The barrel is then boresighted into position.

40.3.1 The velocity-barrel assembly shall be in accordance with the applicable drawings and the following measurements shall be made before it is placed in service, after misfires, pierced primers, flowbacks or whenever a part replacement has been made in the breech block assembly.

	Limits
Firing-pin protrusion	.060" - 0.68"
Firing-pin indent	.011" - .015"
Headspace	.058" - .60"

40.1.4 Lumiline screens are checked for position. It is of the utmost importance that the lumiline screens be placed in their proper position. Measurements must be accurate.

40.2 During firing.

40.2.1 Five(5) warmer (fouling) shots shall be fired. Velocity readings should be recorded by the chronographer to assure that the velocity-measuring equipment is functioning properly.

40.2.2 The recessed holding block containing the required number of standard (reference) cartridges is removed from the controlled temperature room or container and placed at a point convenient to the technician, provided the temperature of the firing room is 70° F ± 5°F. Otherwise the cartridge shall be placed in an insulated box (five cartridges at a time) which has been conditioned at 70°F ± 5°F, and the box placed at a point convenient to the technician. The cartridges are then removed singly from the insulated box immediately before firing. If an insulated box is not available, then the cartridges shall be removed singly from the controlled-temperature room or container immediately before firing.

40.2.3 In order that the propellant shall be uniformly positioned from cartridge to cartridge before firing, attention to detail is necessary in handling and cambering the cartridge.

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cartridge shall be first held vertically, bullet upward, and then rotated slowly,end over end in a vertical plane, stopping the rotation momentarily after 180° of rotation when the bullet is downward, and then continuing through the remainder of 360° , stopping with the cartridge again bullet-end upward. The bullet-end of the cartridge should now be lowered slowly to a position slightly above horizontal. The cartridge shall be chambered very carefully engaged to the hammer. If the technician encounters any difficulty closing the breech-block or engaging the trip lever, the test shall be discontinued until such difficulty is corrected. If any delay should occur after the cartridge is placed in the chamber, and the duration of the delay is approximately 30 seconds or longer, that cartridge shall be extracted and another inserted in its place.

40.2.4 The breech-block shall be closed gently and the trip lever to which the lanyard is attached, shall be carefully engaged to the hammer. If the technician encounters any difficulty closing the breech-block or engaging the trip lever, the test shall be discontinued until such difficulty is corrected. If any delay should occur after the cartridge is placed in the chamber, and the duration of the delay is approximately 30 seconds or longer, that cartridge shall be extracted and another inserted in its place.

40.2.5 The technician retire to a safe position and pulls the lanyard with a smooth firm motion. The velocity of the shock shall be recorded by the chronographer. The breech-block shall be opened, the fired case extracted and visually examined by the technician for possible case casualties.

40.2.6 The procedure prescribed in 40.2.3 through 40.2.5 is repeated until the required number of standards (reference) cartridges have been fired.

40.2.7 The velocity correction shall then be obtained as prescribed in 30.2

40.2.8 Continuous air cooling should be used on the barrel. If air cooling is not available, firing should be regulated so that one cartridge is fired every 30 seconds. The barrel shall be allowed to cool to ambient temperature between each series has been fired. At no time shall the exposed metal surfaces of the test barrel become too hot to grasp with the bare hands (approximately 140°F).

40.2.9 If the average velocity of standard (reference) cartridges is not within ± 35 F/S of the official assessed velocity value, the test barrel shall be removed from the test and another barrel substituted and another series of standards (reference) cartridges fired. If this firing fails to produce satisfactory velocity results, the cause thereof shall be identified and eliminated before resuming the test.

40.2.10 After the barrel has cooled to ambient temperature warming (fouling) shots shall be fired in accordance with 40.2.1.

40.2.11 The test cartridges shall then be fired following the procedure prescribed in 40.2.2 through 40.2.6.

40.2.12 The velocity correction obtained with the standard (reference) cartridges shall then be applied to the mean velocity obtained with the test cartridges as prescribed in 30.2.

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40.2.13 If any of the personnel conducting the test observes any abnormality that may invalidate the velocity measurements, the circumstances shall be reported immediately through appropriate supervisory channels and the test suspended until instructions are received from proper authority. If any result appears abnormal, eve though the possible cause cannot be immediately assigned, the circumstances shall be reported immediately for investigation and decision by proper supervisory authority.

50.0 RECORDING OF DATA

50.1 Results of both standard (reference) and test cartridges, should be recorded directly on the test sheet form.

50.1.1 Standard (reference) cartridges: The individual velocities, average velocity and extreme variation of the individual velocities.

50.1.2 Test Cartridges: The individual velocities, average velocity (not corrected), velocity correction average velocity (corrected) extreme variation and standard deviation of the individual velocities.

50.1.3 Number and type of case casualties.

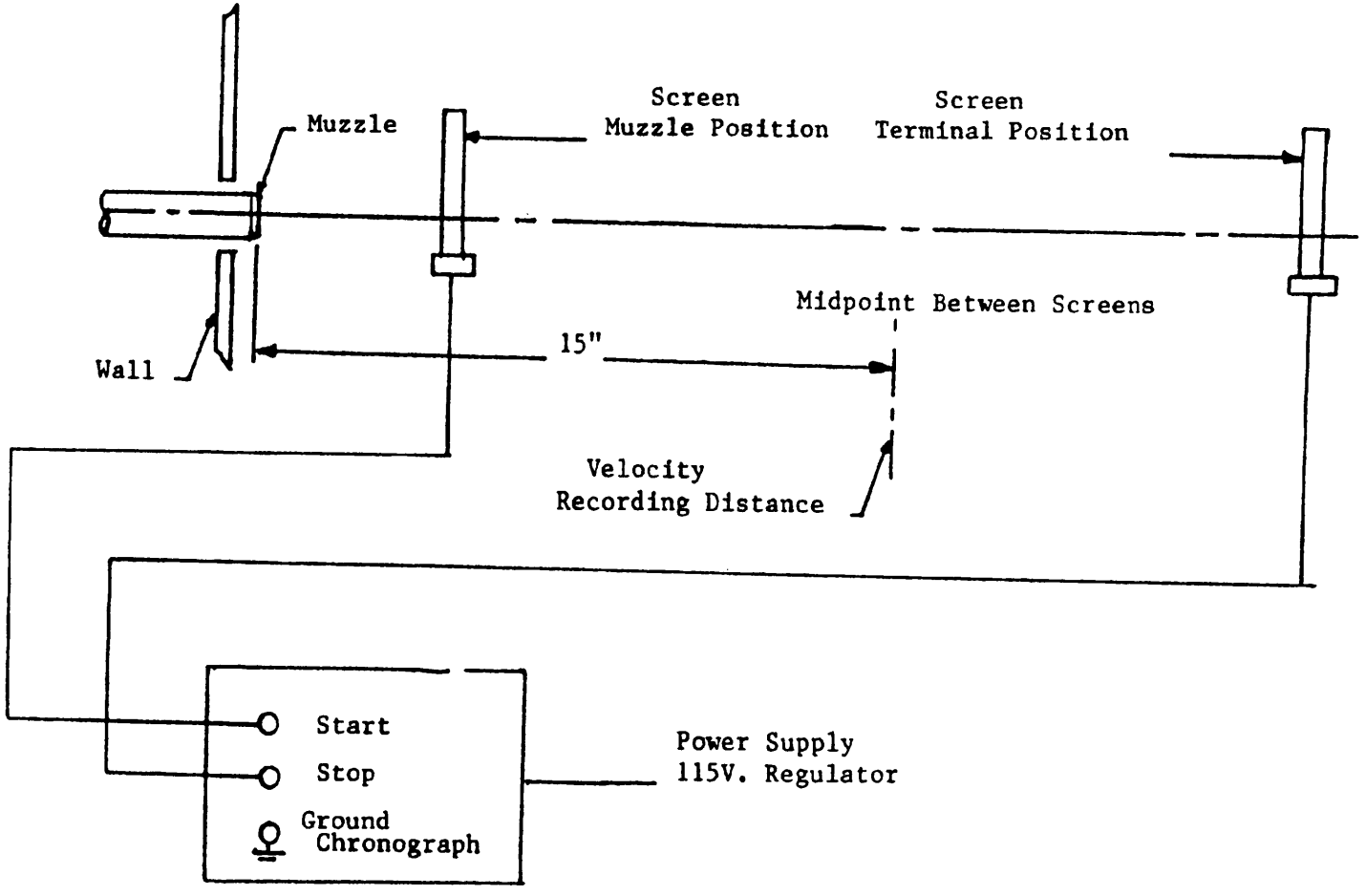
50.2 The following test weapon data shall be recorded on the test sheets.

a. Receiver number.

b. Barrel number.

c. Total number of cartridges fired in barrel,

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Range Equipment, Velocity Test for Bullet Type, Cal. .38

FIGURE C-1

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APPENDIX "D"

CHAMBER PRESSURE TEST PROCEDURE

10.0 PURPOSE

The Chamber-Pressure test is precisely controlled test to determine the chamber-pressure exerted, expressed in pounds per-square inch in the chamber of the gun. The test is performed as a safety measure to insure that the pressure developed by the ammunition is safe for firing in the weapon for which it is intended.

20.0 EQUIPMENT

20.1 Equipment listed in the Chamber-pressure section of the appropriate Inspection Equipment List shall be used.

20.2 The firing range shall be arranged as shown in figure D-1.

20.3 Pier or mount of solid construction on which test fixture assembly shall be mounted.

30.0 USE OF STANDARD (REFERENCE) CARTRIDGES

30.1 Standard (reference) cartridges shall be used to establish range and equipment corrections prior to firing an ammunition lot for acceptance. One standard (reference cartridge shall be fired for each cartridge of the test sample to be fired, up to twenty (20) cartridges. When the test consists of more than twenty (20), but not more than forty (40) cartridges, then twenty (20) standard reference cartridges shall be fired, when the test consists of more than forty (40) cartridges shall be fired, when the one (1) standard (reference) cartridge shall be fired for every two (2) test cartridges.

30.2 After the required number of standards (reference cartridges has been fired, the actual average chamber-pressure of the standard (reference) cartridges shall be compare with the assessed value. If the assessed value is higher than thee actual average chamber-pressure of the standard (reference) cartridges, the difference is a plus correction and shall be added to the average chamber-pressure of the test cartridges. If the assessed value is lower than the average chamber-pressure of the standard (reference) cartridges, the difference is a minus correction and shall be subtracted from the average chamber-pressure of the test cartridges. If the assessed value and the actual chamber-pressure of the standard (reference) cartridges are identical, then no correction is applied.

30.3 The pressure-barrel assembly will be acceptable provided the results obtained with the standard (reference) cartridges are within ± 3500 PSI of the assessed value.

30.4 It is suggested that control charts be maintained for record purposes, of the results from each barrel in which standard (reference) cartridges are fired. Every correction obtained shall be compared with the previous ones to determine whether the test conditions remain stable to a certain degree.

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APPENDIX "D" (CONT'D)

40.0 MEASUREMENT OF COPPER PRESSURE CYLINDERS

40.1 The chamber-pressure shall be determined by using the radial copper pressure cylinders. One cylinder shall be used for each cartridge fired.

40.1.1 The copper pressure cylinder shall be measured individually using a properly calibrated micrometer graduated to 0.0001 inch. The cylinders shall be measured prior to firing, measurements recorded and the cylinders placed in a recessed holding block in such a manner that their identity is maintained throughout the test.

40.1.2 Upon completion of the firing, the cylinders shall be measured again and the decrease in length obtained for each cylinder by subtraction of the actual readings. The decrease in length obtained for each cylinder shall than be applied to the appropriate target table (a target table is supplied with each box of copper pressure cylinders) and the corresponding chamber pressure (PSI) shall be recorded on the test report form in such a manner that the velocity of each cartridge can be identified with the corresponding chamber pressure obtained Velocities recorded at other than ambient are for information only.

40.2 Whenever a new supply of copper pressure cylinders is received involving the use of different target tables, they shall be tested in comparison with the cylinders being replaced by firing forty-four (44) standard (reference) cartridges for chamber pressure in a barrel assembly of know characteristics, using twenty-two (22) of the new cylinders in alteration with twenty-two (22) of the old cylinders shall not vary by more than ± 800 PSI from the mean obtained with the old cylinders. If this limit is exceeded, the new lot is considered unacceptable and reported to the responsible agency.

40.3 TEST PROCEDURE

40.3.1 Pre-firing (Preparation for test)

40.3.1.1 The required number of test and standard (reference) cartridges shall be placed in a vertical position primer end down, in recessed holding blocks.

The cartridges shall be permitted to come to a temperature of 60° and 80°F prior to being placed in the temperature-controlled room or container in such a manner that all cartridges are subjected to a uniform temperature for a minimum of two hours, prior to firing. The container or room shall be maintained at a temperture of 70°F+2° F, with a relative humidity of 60 percent, ± 5 percent, and be sufficient capacity to allow free circulation of air.

40.3.1.2 The pressure-barrel assembly is assembled in the test fixture on the mount. The chamber and bore of the barrel are wiped dry. The barrel is then boresighted into position.

40.3.1.3 The following measurements shall be made before a barrel assembly placed in service, after misfires, pierced primer, flowbacks or whenever a part replacement has been made in breech-block assembly.

	Limits
Firing-pin-indent	.011" - .015"
Firing-pin protrusion	.060" - .068"
Headspace	.058" - .060"

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40.3.1.4 Lumiline screens are checked for position. It is of utmost importance that the lumiline screens be placed in their prior positions. Measurements must be accurate.

40.3.1.5 The required number of copper pressure cylinders is measured as prescribed in 40.1.1.

40.3.1.6 The recessed holding block containing the individually measured copper pressure cylinders, a sufficient quantity on obturating cups and the pressure piston prefixally fitted to the barrel assembly should be placed at a point convenient to the technician.

40.3.1.6.1 The obturating cups (wax filled) are not changed after firing each shot. It is permissible to fire a maximum of twenty-two (22) cartridge with the same obturating cup. The first two cartridges are fired to properly seat the obturating cup in the piston hold, the results obtained with these two cartridges are recorded for information only and are not considered in the final averages.

40.3.2 During firing

40.3.2.1 Five (5) warming (fouling) shots shall be fired. To fire that warming (fouling) shots,, it shall be necessary to service the chamber-pressure barrel assembly with an obturating cup, the pressure pistol and a copper pressure cylinder. The same cylinder and obturating cup may be used when all the warming shots are being fired. The anvil is screwed down on the cylinder following each shot. Velocity readings should be recorded by the chronographer to assure that the velocity-measuring equipment is functioning properly.

40.3.2.2 The recessed holding block containing the standard (reference) cartridges is removed from the controlled-temperature room or container and placed at a point convenient to the technician, provided the temperature of the firing room is 70°F ± 5°F. Otherwise the cartridges shall be placed in an insulated box which has been conditioned at 70°F ± 5°F, and the box placed at a point convenient to the technician. The cartridges are then removed singly from the insulated box immediately before firing. If an insulated box is not available, then the cartridges shall be removed singly from the controlled-temperature room or container immediately before firing.

40.3.2.3 The dummy cartridge is placed in the chamber of the barrel assembly. The obturating cup shall be placed in the pistol hole, mouth-end down, and partially seated using the stem of the knockout tool. The shank of the pressure pistol shall be dipped in oil, SAE40 to equivalent,and the oil then be allowed to drawing from the piston. The drop of oil adhering to the bottom of the piston shank shall then be removed by touching the bottom of the the pistol shank to a cleaning patch. The piston shall ben be inserted into the pistol hole and pressed down on the obturating cup until the pistol has reached its correct final position. The head of the piston and the bottom of the anvil shall be wiped dry and free of oil. The copper cylinder shall be put in place and centered between the head of the piston and the bottom of the anvil. The anvil shall be screwed down lightly on the cylinder, using thumb and forefinger. but not under stress. the dummy cartridge shall then be removed from the chamber. Chamber and bore are examined for possible obstructions.

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40.3.2.4 In order that the propellant shall be uniformly position from cartridge to cartridge, attention to detail is position from cartridge to cartridge, attention to detail is necessary in handling and cambering the cartridge. The cartridge shall first be held vertically, bullet upward, and then rotated slowly, end over end in a vertical plane, stopping the rotation momentarily after 180° of rotation when the bullet is downward, and then continuing through the remainder of 360 °stopping with the cartridge again bullet-end upward. The bullet-end of the cartridge should now be lowered slowly to a position slightly above horizontal, The cartridge shall be chambered very carefully, taking care that the primer-end of the case is not elevated above the bullet-end. (The object is to have the cartridge seated in the chamber ready to fire, with the propellant in a loose condition at the primer-end of the ease).

40.3.2.5 The breech-block shall be closed gently. If the technician encounters any difficulty closing the breech-block or engaging the trip lever, the test shall be discontinued until such difficulty is corrected. If any delay should occur after the cartridge is placed in the chamber and the duration of the delay is approximately 30 seconds or longer, that cartridge shall be extracted and another inserted in its place.

40.3.2.6 The technician makes a final check to assure that the anvil is screwed to a snug position on the copper cylinder. Care is taken to see that the copper cylinder is not compressed by the anvil prior to firing. The proper torque to be applied to the thumscrew is about on pound-inch. (The torque can be estimated with satisfactory accuracy by an experienced technician; to familiarize inexperienced technicians with the desire degree of tightness, a torque measuring device may be employed. This can be accomplished by drilling and tapping (threading) an axial hole in the knurled head of the anvil (thumb-screw) and inserting a bolt or screw to which a torque-measuring wrench can be attached. The usefulness of the anvil (thumb-screw) is not impaired by this modification).

40.3.2.7 The trip lever, to which the lanyard is attached, shall be engaged gently to the hammer. The technician retires to a safe position and pulls the lanyard with a smooth firm motion. The velocity of the shot shall be recorded by the chronographer.

40.3.2.8 The breech-block shall be opened, the fired case extracted and visually examined by the technician for possible case casualties. The copper cylinder shall be removed and placed in the recessed holding block. (Because the same obturating cup is used to fire twenty-two (22) cartridges, it is not necessary to remove either the obturating cup or type piston from the piston hole from shot to shot). The piston is then pressed down in the piston hole to is correct final position; this can be accomplished by gently tapping the head of the piston with the knockout tool or by placing a copper cylinder on the head of the piston and screwing the anvil down on it forcing the piston into its correct final position. The cylinder used for this purpose in the removed from the head o the piston and replaced with the next cylinder in the recessed holding block. The anvil shall be screwed down lightly on the cylinder, using thumb and forefinger, but not under stress. The chamber and bore of the test weapon are checked for obstructions.

40.3.2.9 The procedure as specified in 40.3.2.4 through 40.3.2.8 is repeated until the twenty-two (22) cartridges have been fired.

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40.3.2.10 The piston is then removed from the piston hole. The knockout tool is then used to force the obturating cup from the piston hole into the chamber. The obturating cup is then removed from the chamber using a cleaning rod with a dry cleaning patch attached.

40.3.2.11 Continuous air cooing should be used on the bard; however, if air cooling is not available, firing should be regulated so that one cartridge is fired every thirty (3) seconds. At no time shall the exposed metal surface of the test barrel become too hot to grasp with the bare hands (approximately 140°F).

40.3.2.12 The copper pressure cylinders whose identities are maintained throughout the test are measured in the same manner as they were prior to the test. The difference in length (set in inches) is then applied to the primer tarage table, and the corresponding PSI is obtained and recorded on the report form. The chamber pressure correction shall then b obtained as specified in 30.2.

40.3.2.13 If the average chamber pressure of the standard (reference) cartridges is not within ± 3500 PSI of the official assessed value, the test barrel shall be removed from the test, another test barrel substituted and another series of standard (reference) cartridges fired. If this firing fails to produce satisfactory results, the cause thereof shall be identified and eliminated before resuming the test.

40.3.2.14 After the barrel has cooled to ambient temperature, warming (fouling) shots shall be fired in accordance with 40.3.2.1.

40.3.2.1.5 The test cartridges shall then be fired following the procedure as specified in 40.3.2.2 through 40.3.2.9.

40.3.2.16 The copper cylinders used with the test cartridges are measured in the same manner as they were prior to the test. The chamber-pressure readings shall be placed on the test sheet in such a manner that the velocity and chamber-pressure of each shot is identifiable. The chamber-pressure correction obtained with the reference cartridges shall be applied to the average chamber-pressure of the test cartridges, as specified in 30.2.

40.3.217 If any of the personnel conducting the test observes any abnormality that may invalidate the velocity or pressure measurements, the circumstances shall be reported immediately through appropriate supervisory channels and the test suspended until instructions are received from proper authority. If any result appears abnormal, even though the possible cause cannot be immediately assigned, the circumstances should be reported immediately for investigation and decision by proper supervisory authority.

50.0 RECORDING OF DATA

50.1 Results of both standard (reference) cartridges and test cartridges should be recorded directly on the test sheet form. Velocities shall be recorded to the nearest FPS; chamber-pressure shall be recorded to the nearest 100 PSI.

50.1.1 Standard (reference) cartridges - The individual velocities and chamber-pressures, average velocity and chamber-pressure and extreme variations of velocity and chamber-pressure shall be recorded.

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APPENDIX "D" (CONT'D)

50.1.2 Test cartridges - The individual velocities and chamber-pressures, average velocity and camber pressure (not correct), chamber-pressure correction, average chamber-pressure (corrected), extreme variation of the individual velocities and the individual chamber-pressure shall be recorded.

50.1.3 Number and type of case casualties.

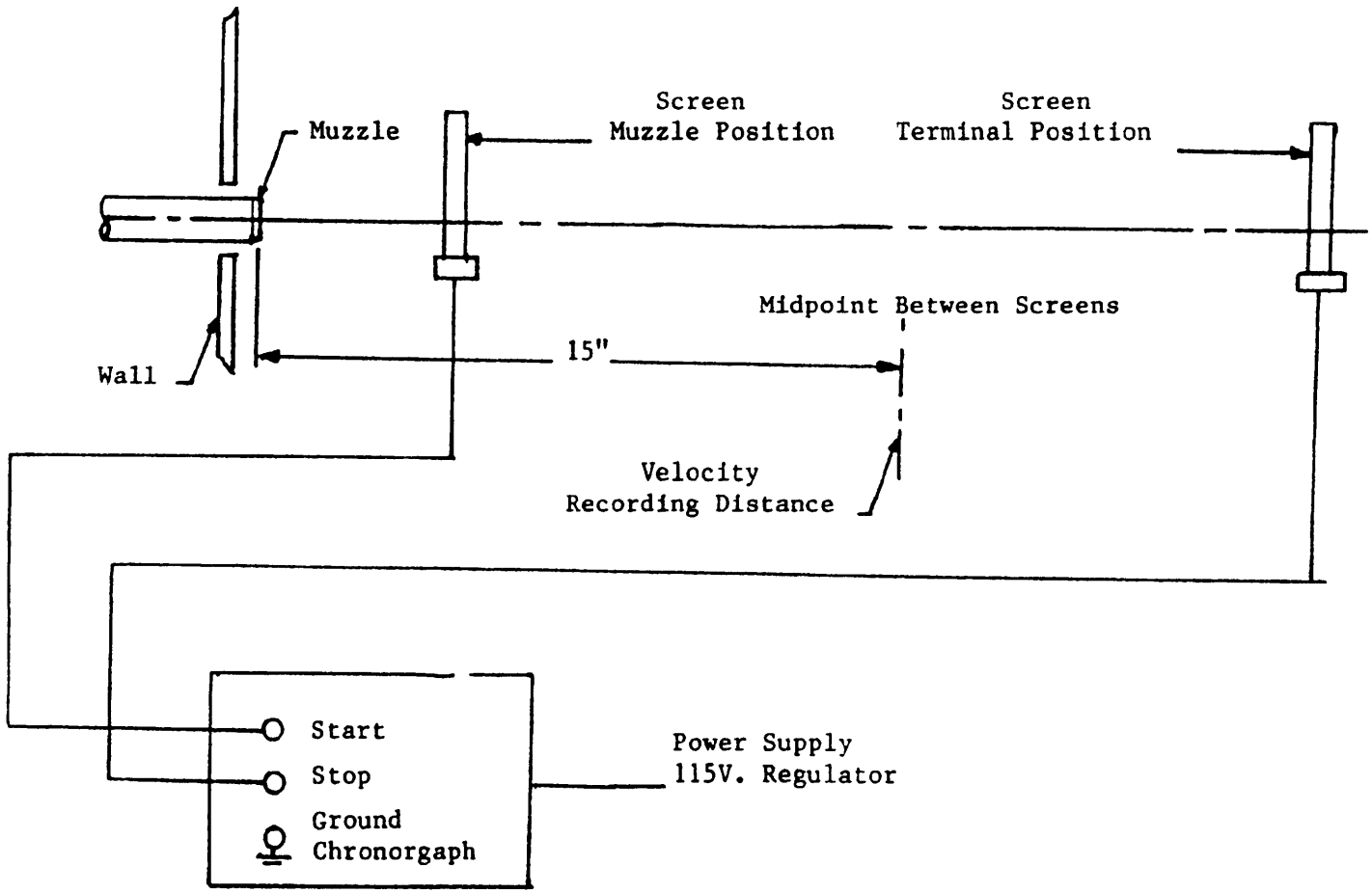
50.2 The following test weapon data shall be recorded on the test weapon data shall be recorded on the test sheet:

a. Receiver number.

b. Barrel number.

c. Total number of cartridges fired in barrel.

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Range Equipment, Velocity Test for Bullet Type, Cal. .38

FIGURE D-1

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APPENDIX "E"

ACCURACY TEST PROCEDURE

10.0 PURPOSE

The accuracy test is fired under controlled conditions at a target located at a Specified distance from the weapon to determine the uniformity and dispersion of bullets. To minimize weapon inaccuracies, the test shall be fired in rigidly mounted test weapon.

20.0 EQUIPMENT

20.1 Equipment listed in the Accuracy section of the appropriate Inspection Equipment List shall be made.

20.2 The target size for accuracy tests shall be not less than 24" x 24".

20.3 The pier or mount on which the test assembly is mounted shall be of solid construction.

30.0 TEST PROCEDURES

30.1 Pre-firing (preparation for test)

30.1.1 The accuracy test weapon is assembled and locked in position on the mount end foresighted into position. It is of prime importance that the assembly be mounted properly is that the weapon maintains its original position from shot to shot.

30.1.2 The following measurements shall be made before a test weapon is placed in service, after misfires, pierced primers or flowbacks; or whenever a part replacement has been made in the assembly:

Firing-pin	.060"- .068"
Firing-pin indent	.011"- .015"
Headspace	.058"- .060"

30.1.3 The test cartridges shall e placed at a point convenient to the technician. It is not necessary to condition the ammunition at a specified temperature prior to firing.

30.1.4 At least two (2) but preferably three (3) weapons shall be used. The number of targets to be fired shall be divided among the number of weapons to be used.

30.2 During Firing

30.2.1 A sufficient number of unrecorded cartridges the type of ammunition under test shall be fired to assure that the test weapon is correctly sighted on the target, but in any even, a minimum of three (3) cartridges shall be fired to warm and foul the weapon when it is first put into service and after it has been cleaned or cooled. The technician observes the location of the shots on the target so that the alignment of the weapon may be adjusted, if necessary.

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30.2.2 After the warming (fouling) cartridges have been fired the target shall be changed so as to present a fresh surface for the succeeding target. Thereafter, the target shall be changed after each target often cartridges has been fired.

30.2.3 Consideration is not given to the position of the propellant in the cartridge case,except that the reamer of handling the munition from cartridge to cartridge is reasonably uniform. Ten (1) cartridges shall then be fired in even sequence at a rhythmic uniform rate,as rapidly as service of the weapon permits.

30.2.4 The approximate location of the center of impact and the estimated size (Extreme Vertical and Extreme Horizontal) of the target first fired shall be observed by the technician in order that alignment of the weapon may be adjusted, if necessary.

30.2.5 The procedure prescribed in 30.2.3 and 30.2.4 shall than be repeated until the specified number of targets have been fired in the first weapon, five (5) targets maximum, or when the exposed metal surface of the test barrel becomes too hot to grasp with the bare hands (approximately 140° F), at which time the barrel shall be cooled to ambient temperature before the test is continued.

30.2.6 After the required number of targets has bee fired with the first weapon, that weapon shall be removed from the mount and cooled to ambient temperature.

30.2.7 The second test weapon shall then be assembled on the mount and the procedure prescribed in 30.2.1 through 30.2.6 is repeated. If a third test weapon is to be used, the same procedure shall be repeated

30.2.8 If it is necessary to use a weapon after it has been cooled, the chamber and bore shall be cleaned before additional firing, and the three warning (fouling) cartridges shall be repeated.

30.2.9 The technician shall record all pertinent information relative to the test, i.e., misfires, failure to trace, case casualties,etc. Whenever the validity of the test is in doubt, test equipment can be checked by firing ammunition of known accuracy.

30.3 MEASUREMENT OF TARGET

30.3.1 Explanation of terms.

30.3.1.1 Diagonal (D). After drawing a rectangle whose four lines pass through the outermost shot-holes on a target, the diagonal (hypotenuse) is measured to the nearest tenth of an inch.

30.3.1.2 Mean Diagonal (M.D.) The arithmetic mean of all diagonals.

30.3.1.3 Figure of Merit (F. M.) The arithmetic average of the Extreme Vertical and he Extreme Horizontal.

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30.3.1.4 Mean Radius (M.R.) The arithmetic mean of the distances between the center of all shot-holes and a point of the target called the Center of Impact.

30.3.1.5 Extreme Vertical (Ex. Ver.). Vertical distance between the center of the hole made by the uppermost shot and the center of the hole made by the lowermost shot.

30.3.1.6 Extreme Horizontal (Ex. Hor.) Horizontal distance between the center of the hole made by the shot farthest to the right and the center of the hole made by the shot farthest to the right and the center of the hole made by the shot farthest to the left.

30.3.1.7 Extreme Spread (Ex. Spr.). Distance between centers of the shot-holes farthest apart.

30.3.1.8 Center of Impact (C.I.). Defined as the point at which the algebraic sum of the components; of the distances to the center of each shot-hole is zero.

30.3.2 The procedure for obtaining the Diagonal (D) for each target shall be as follows:

30.3.2.1 One each target, draw a rectangle whose four sides pass through the centers of the out most holes (See Figure E-3). The top and bottom lines of the rectangle shall be horizontal; the vertical lines shall be perpendicular to the horizontal lines.

30.3.2.2 The Diagonal (D) of each rectangle shall then be measured to the nearest tenth of an inch. The Diagonal may be measured from the upper left comer to the lower right corner, or from the upper right comer to the lower left comer.

30.3.2.3 The Mean Diagonal (M.D.) shall be obtained by adding the individual diagonal measurements and dividing by the number of targets.

30.3.3 Mean Radius for targets shall be determined in the manner indicated below, and illustrated in Figures E-1 and E-2.

30.3.3.1 Draw a horizontal line, PA, through the center of the lowest shot-hole on the target and extend it horizontally in both directions until its length is equal to or greater than the width of the group of shots comprising the target.

30.3.32 Draw a vertical line, PB, through the center of the shot-hole on the extreme left of the target perpendicular to and meeting line PA at P. Extend line PB above PA unil the length of PB is equal to or greater than the height of the group of shots comprising the target.

30.3.3.3 Measure (by means of the scale or opisometer) the perpendicular distance Y from the center of each shot-hole on the target to line PA. Add the distances so obtained d divide by the number of shots. The results of this calculation is the average vertical distance of the shot-holes above the base line PA and is designated by Y. Similarly, measure the perpendicular distance X from the center of each shot-hole to line PB, add the distances so obtained and divide by the number of shots. The results is the average horizontal distance of the shot-holes from the vertical ordinate PB, and is designate X.

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30.3.3.4 Location of Center of Impact. From P, measure along PB a distance, PD, equal to Y and along PA measure a distance, PE, equal to X. At points E and D erect the respective perpendiculars EC and DC to lines PA and PB. The intersection of these perpendiculars is the Center of Impact of the target, C. Steps 30.3.3.1 and 30.3.3.4 inclusive, are illustrated in Figure E-1.

30.3.3.5 Mean Radius (M.R.) From the Center of Impact, measure the distance to the center of each shot-hole of the target Add these distances and divide by the number of shot-holes, the result is the M.R. Figure E-2 illustrates this procedure.

30.3.3.6 If one of more shots miss the target in any ten shot series, the entire accuracy test shall be considered invalid and refired at a larger target, as necessary to ensure hits.

40.0 RECORDING OF DATA

40.1 All tests results shall be recorded to the nearest one-tenth of an inch.

40.2 Test sheets shall show the following:

a. Results of all specified requirement.

b. Number of type of case casualties.

c. Misfires.

d. Stripped bullets.

40.3 The following test weapon data shall be recorded on the test sheets:

a. Receiver number.

b. Barrel Number

c. Total number of cartridges fired in barrel.

d. Headspace measurement.

40.4 FACTORS AFFECTING ACCURACY TESTS

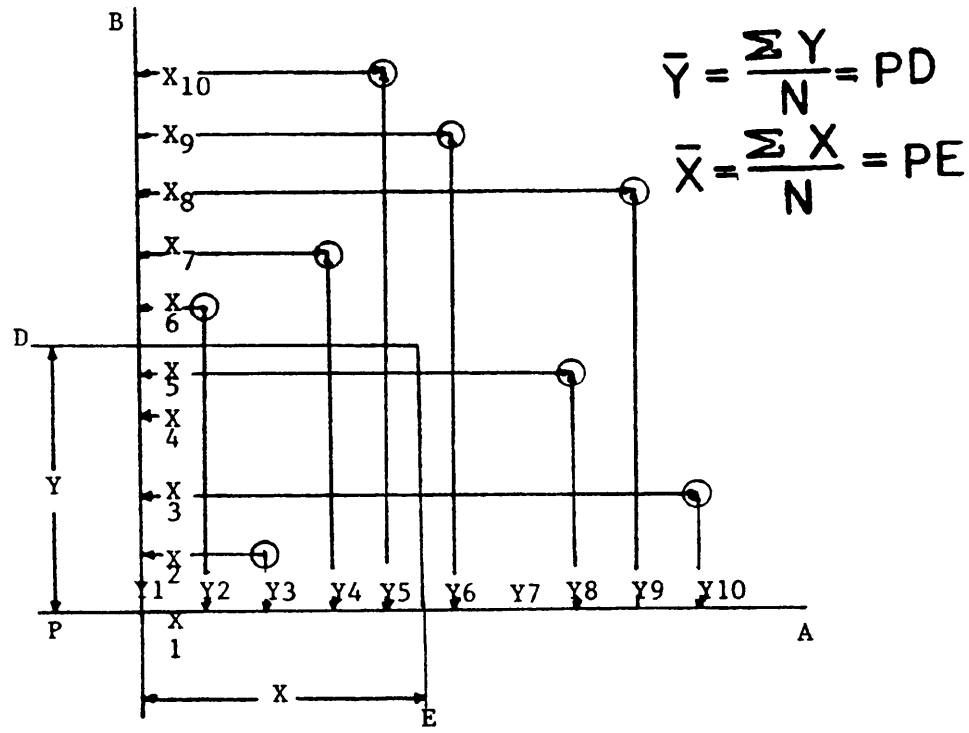
40.4.1 The external factors should be controlled as closely as possible in order to obtain results that are representative of the inherent accuracy of the ammunition.

40.4.2 The dimensions and condition of accuracy weapons, as well as the manner of placing the weapon the rest, are of prime importance.

40.4.3 The technique of test can affect accuracy results. If reproducibility in the manner in which the cartridges are handled, chambered and fired be poor, a larger Mean Diagonal or Mean Radius is obtained than if the cartridges are tested in a uniform manner.

40.4.4 The temperature of the test barrel should be controlled so that the exposed metal surface of the barrel does not become too hot to grasp with the bare hands (approximately 140 °F), at which time the weapon shall be cooled to ambient temperature before the test is continued.

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E-1

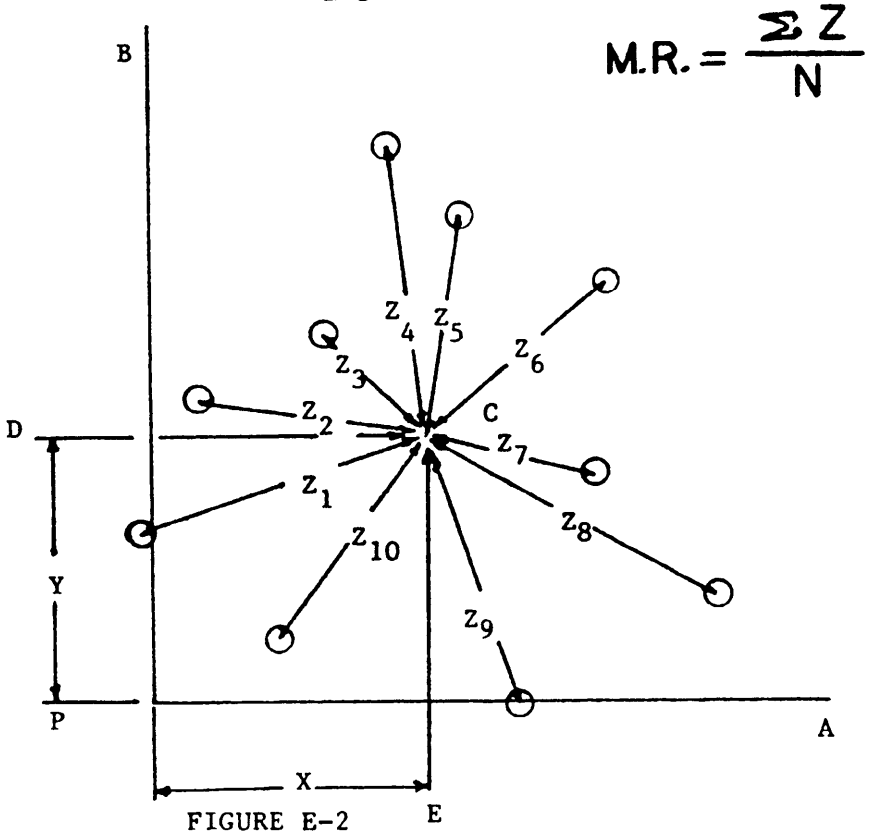


FIGURE E-2

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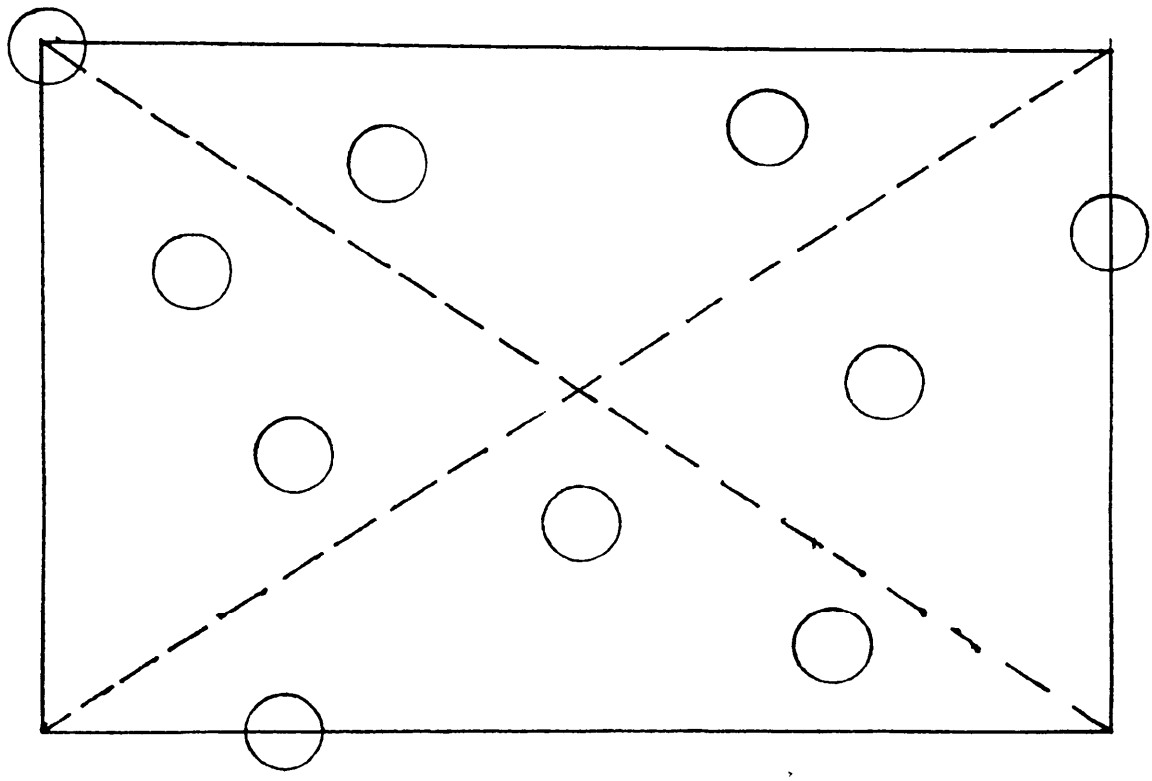


FIGURE E-3

MEAN DIAGONAL

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APPENDIX "F"

FUNCTION AND CASUALTY TEST PROCEDURE

10.0 PURPOSE

To determine if the ammunition can be expected to perform satisfactorily in the service weapons for which it has been designed under conditions of field usage. Ammunition may be ballistically satisfactory, i.e., it may comply with individual performance specifications such as velocity, pressure, etc., yet be unfit for use in the field because of undesirable characteristics which jeopardize the safety of personnel.

Casualties and malfunctions can be caused wither by the ammunition or by the weapon in which it is fired,so, that to a certain extent, these two factors are interdependent. A faculty weapon or a poorly adjusted weapon can cause casualties in normal casualties are encountered, the the fault lies with the ammunition.

20.0 EQUIPMENT

Equipment listed in the Function and Casualty section of the appropriate inspection Equipment List shall be used.

30.0 TEST PROCEDURE

30.1 Pre-firing (Preparation for test)

30.1.1 Weapons shall be of the latest design or most recent issue. Weapons shall be discarded when unusual behavior occurs which is proven attributable to the weapon.

30.1.2 Weapons employed in this test must have bards in serviceable condition. To assure this, all barrels shall be inspected visually before use and the weapon rejected if any defect is discovered. Unusual behavior, proven to be attributable to the barrel, may also be reason for discarding the weapon.

30.1.3 The test cartridge shall be examined for obvious defects before the Function and Casualty test. If visual defects are found, the defective cartridge (s) shall be replaced and the defects shall be noted on the test sheet form.

30.1.4 The test weapon may be fired off-hand or in a fixed rest, as desired. Firing shall be alternated, i.e., six (6) cartridges single action, six (6) cartridges double action, six (6) cartridges single action, etc.

30.2 During firing.

30.2.1 The number of cartridges fired single and double action (6 each) shall be governed by the capability of the cylinder of the revolvers. The weapon shall be at room temperature at the beginning of the test, and shall be cooled after firing 48 cartridges.

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30.2.2 The number of revolvers to be used and the quantity of cartridges to be fired in each, be as specified in the applicable specification. The procedure for firing each of the weapons specified shall be the same insofar as possible. Six (6) cartridges shall be inserted into the cylinder of the weapon and fired single action; 6 six cartridges are then inserted and fired double action; this procedure is followed until the specified number of cartridges have been fired. A time interval of not more than one minute shall be allowed to reload the weapon each time.

30.2.3 Fired cases shall be visually examined by the technician for possible case casualties.

40.0 RECORDING OF DATA

40.1 Casualties shall be reported in accordance with the terminology specified in 60.0 and the applicable specification.

40.2 Misfires shall be recorded and the cause described (see 50.2)

40.3 The Function and Casualty Test requires careful attention and alertness, and any unusual occurrence in the weapon function or appearance of fired cases shall be noted.

40.4 Failure of the gun parts shall be shown on the ammunition report.

50.0 OPERATIONAL NOTES

50.1 In the event any stoppages occurs during firing of the test, a detailed check shall be made to determine whether ammunition or the equipment is a fault. If the stoppage was caused by a misfire, the weapon shall be checked thoroughly. To assist in determining whether ammunition or equipment is responsible for a stoppage, it is good practice to test the weapon in question using ammunition of know characteristics, and to test the ammunition in question by firing in another weapon of the same type. It it is established that some faulty condition of the weapon is responsible for the stoppage, the test shall be disregarded, the weapon shall be corrected or replaced and the tests with that type weapon shall be refired.

50.2 If a misfired is encountered, irrespective of the type of test wherein it occurs, the weapon is examined carefully to determine if the cause is attributable to the weapon. In any ballistic acceptance test where a misfire occurs, a second attempt to fire a primer shall not be made. It is mandatory that a period of at least five (5) minutes elapse after the misfire occurs before removing the cartridge from the weapon. Existing safety regulations shall be exercised when removing the cartridge from the chamber; the cartridge is then preserved for further examination. All handling and examinations of misfired cartridges shall be conducted with due regard for hazards involved. The weapon in which a misfire occurs shall be checked thoroughly; it shall be disassembled and all component parts critically scrutinized. Results of such examination shall be included on the rest report as a matter of information.

50.2.1 Laboratory examination of the misfired cartridges shall be made to determine the specific cause; the result of the investigation is include on the test report.

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50.3 Upon completion of firing, all cartridge cases from the test ammunition shall be carefully examined for firing defects. If any defect is found, a detailed check of the equipment shall be made to determine whether the ammunition or the equipment is at fault. If it is established that a faulty weapon is responsible for the firing defect, than the test shall be disregarded, the weapon shall be corrected or replaced and the tests with that one weapon shall be refired. If it cannot be established that the weapon or other equipment is at fault, then the firing defects shall be charged against the ammunition.

60.0 DEFINITIONS

60.1 Misfire. Failure of a round or ammunition to fire after the initiation impulse has bee applied to the primer.

a. The primer fails to fire when struck by the firing pin.

b. The propellant does not ignite when the primer fires normally.

60.2 Hangfire. Any delay in the functioning of a cartridge after the initiating impulse has been applied to the primer.

60.3 Pierced primer. A pierced primer is one in which the primer cup is entirely perforated by the firing pin. It can be identified by a visible hole through the primer, or if the perforation be minute, by discoloration of the firing pin indent caused by burning gas. This casualty may be caused by an imperfect firing pin or by one of improper contour; or it may be caused by the metal in the base of the primer cup being too thin, or too soft, or defective from lamination, scale etc.

60.4 Primer leak. Discoloration caused by gas leakage around the junction between the primer cup and the primer pocket wall.

60.5 Loose primer. Looseness which permits the fired primer to fall from the primer pocket after the cartridge is fired, without distortion to the primer pocket.

60.6 Blown primer. A blown primer is a primer which,m when the cartridge is fired, is separated completely from the head of the cartridges case, and both the head of the case and pocket are enlarged and deformed. This condition results from a soft head.

60.7 Flow-back. A flow-back may be described as the distortion al foxing of the dome of the primer cup in the the space between the firing pin and the firing pin hole in the bolt face.

60.8 Punch-out. A punch-out is the removal of metal in the dome of the primer.

60.9 Primer setback. The backward movement of a primer cup in the cartridge case which occurs when the base of the cup is not properly supported by the bolt face, and protrudes above the head of the cartridge case.

60.10 Ruptured case. Circumferential separation of the case wall produced by firing. Ruptures are divided into two categories, partial and complete. A partial rupture is one which

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extended less then 360 degrees around the case a complete rupture is on which extended entirely around the case, separating the case into two parts. Ruptures are designated according to position, as indicated on Drawing B7643674.

60.11 Split ease. A longitudinal separation of the metal in the case wall produced by firing. Splits shall be classified as prescribed by the cartridge specification and Drawing C7643674.

60.12 Rim shear. A case which exhibits complete removal of any portion of the extractor rim during normal gun functioning, usually resulting in failure to extract.

60.14 Failure to extract. The fired case is not removed from the weapon chamber by normal gun action.

60.15 Bullet-in-bore. A bullet remaining in the bore of the bad.

60.16 Muzzle burst. Explosion of a projectile at the muzzle of the weapon, or up to a distance of approximately fifteen (15) feet from the muzzle.

60.17 Muzzle flash. Flash that appears at the muzzle of the weapon when a projectile leaves the barrel.

60.18 Breech flash. Gas flash occurring at breech of weapon.

60.20 Fouling Deposit that accumulates in receiver of rifle or bore of barrel during firing.

60.21 Dropped primer. A primer which, when the cartridge is fired, falls from the primer pocket during or after extraction. Although it is in the same category as a blown primer, deformation to the head of the case and the primer pocket is less obvious.

60.22 Stripped jacket. The jacket of the bullet separates from the core after it has left the muzzle of the weapon or separates the bore.

60.23 Stretched ease. A longitudinal elongation of the metal of the case wall which can be recognized by a change in sheen of the case within a partial or complete circumferential ring or band:

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APPENDIX "G"

TEMPERATURE (HIGH AND LOW) TEST PROCEDURE

10.0 PURPOSE

To determine the ballistic stability of ammunition after being stored at high or low temperature.

20.0 EQUIPMENT

20.1 Equipment listed in the Temperature Test section of the appropriate Inspection Equipment List shall be used.

20.2 Temperature-controlled container or containers capable of maintaining temperature within a tolerance of ± 2°F (insulated boxes which have been brought to the required temperature may be used to transfer ammunition from the controlled-temperature storage to the test weapon if portable controlled-temperature containers are not available).

20.3 The temperature-controlled container(s) shall be of sufficient capacity to permit conditioning the ammunition to the desired temperature, and allow free circulation of air around the cartridges.

30.0 TEST PROCEDURES

30.1 Pre-firing (Preparation for test)

The same procedures used in preparation for firing Chamber Pressure (Appendix "D") and Function and Casualty (Appendix "F") shall be used for this test.

30.1.1 For purpose of safety, the Chamber Pressure test shall be conducted prior to firing the Function and Casualty Test. If the chamber pressure of any individual cartridges exceeds the limit set in the applicable specification, the Function and Casualty Test shall not be conducted.

30.1.2 Temperature of test ammunition shall be 60° to 80°F, prior to conditioning at high or low temperatures. the ammunition shall be placed in the temperature-controlled containers(s) in such a manner that all cartridges are subjected to a uniform temperature, and maintained for the time specified.

30.2 During firing.

30.2.1 The ammunition under test, stored at the temperature specified in the applicable specification is removed from the controlled temperature container, and placed in the insulated box (referenced in 20.2), which has also been conditioned to the temperature specified. The insulated box is then placed at a convenient point the technician.

30.2.2 The procedures used for firing Chamber Pressure Test (Appendix "D") and Function and Casualty Test (Appendix "E") shall be used for this test with the following exceptions:

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APPENDIX "G" (CONT'D)

For chamber Pressure firing, three cartridges are removed from the controlled-temperature container and placed in the insulated box. The cartridges are then removed singly from the insulated box and fired.

For Function and Casualty firing, six cartridges are removed from the controlled temperature container and placed in the insulated box. The cartridges are then loaded in the revolver and fired as specified.

These procedures shall be followed for either high or low temperature tests.

40.0 RECORDING OF DATA

40.1 Results shall be recorded directly on the test report form as follows:

40.1.1 Chamber pressure. Individual velocities and chamber pressures, average velocity and chamber pressure, chamber pressure correction, average chamber pressure (corrected), standard deviation of the individual chamber pressure.

40.1.2 Function and Casualty. Number and type of case casualties and weapon malfunction.

40.1.3 Temperature at which test cartridges were stored and fired.

40.1.4 The following test weapon data shall be recoded on the test report form:

a. Receiver number.

b. Barrel number.

c. Total number of cartridges fired in the bard.

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APPENDIX "H"

WATERPROOF TEST PROCEDURE

10.0 PURPOSE

To determine the water tightness of cartridges.

20.0 EQUIPMENT

The equipment shall be the equivalent of that shown on drawing listed in the Waterproof section of the appropriate Inspection Equipment List.

30.0 TEST PROCEDURES

30.1 Preparation for test.

30.1.1 Convenient configuration for the reservoir and glass test chamber are shown on the drawing referenced in 20.0. the glass test chamber should be at least of sufficient diameter to accommodate five cartridges lying horizontally on a perforated rack or tray and of sufficient depth to allow at least two inches of water above the cartridges for the test. The volume of the reservoir should be about five to fifteen time the volume of air remaining in the glass test chamber when the desiccator contains sufficient water for conducting the test.

30.1.2 The glass test chamber shall contain a sufficient amount of freshly boiled water (cooled at room temperature) to allow 2 to 2-1/2 inch head of water above the cartridge. A perforated metal tray shall be placed across the narrow part of the body. The ground glass surfaces on the lid and body and on the glass top shall be smeared with vaseline or vacuum grease.

30.2 Conducting the test.

3.2.1 With the glass test chamber closed to the vacuum reservoir and the valve open from the reservoir to the vacuum pump, the vacuum pump shall be set in operation. The pressure in the reservoir shall be reduced until the vacuum gage shows some predetermined reading greater than that specified in the specification; this predetermined reading is the vacuum in the reservoir which will, when the reservoir is connected to the glass test chamber, produce the desired vacuum in both. (The desired vacuum in the reservoir will be approximately one to three inches greater than that required by the specification and should be determined by experiment with the particular apparatus to be used, and recoded for future reference). When the desired vacuum in the reservoir has been obtained, the valve from the reservoir to the vacuum pump shall be closed, and the pump operation may be stopped.

30.2.2 The ammunition to be tested (not exceeding the number specified) shall be placed horizontally on the tray in the glass test chamber and the lid placed in position. The glass tap shall be turned to allow the vacuum resevoir to evacuate the glass test chamber to the

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required pounds per square inch below atmospheric pressure and shall be held at that pressure for the specified time. The number of bubbles liberated from the mouth or primer, or both, of each cartridge shall be observed. At the end of the specified time, the vacuum shall be released from the glass test chamber, the lid removed and the ammunition removed.

30.2.3 the procedure prescribed in 30.2.1 and 30.2.2 shall be repeated until the required number of cartridges has been tested.

40.0 RECORDING OF DATA

Results of Waterproof Test shall be recorded as follows:

40.1 No leak.

40.2 slow leak. A series of two or more air bubbles appearing at the primer or mouth of the case, or both but which are liberated at such rate that only one bubble from either the primer or the mouth of the cartridge is in transity to the surface at any one time.

40.3 Fast leak. A series of air bubble appearing at the primer of mouth of the case, or both, which are liberated at such a rate that more than one bubble from the primer or the mouth of the cartridge case is in transit to the surface at any one time.

40.4 Location of leak. Identify area of leak with machinist layout ink.

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APPENDIX "I"

CARTRIDGE ELONGATION TEST PROCEDURE

10.0 PURPOSE

When firing ammunition in a revolver, the recoil of the weapon ends to cause the bullets in the unfired cartridges to move forward causing the cylinder to jam, preventing it from revolving. The Cartridge Elongation test is fired to determine if the bullets contained in the test ammunition are held in the cartridge cases in such a manner to prevent their forward movement while other cartridges are being fired.

20.0 EQUIPMENT

20.1 Equipment listed in the Cartridge Elongation section of the appropriate Inspection Equipment list.

20.2 Firing range with suitable bullet-stop.

30.0 TEST PROCEDURES

30.1 Pre-firing (Preparation for test)

30.1.1 The number of cartridges as prescribed in the applicable specification are placed in a recessed holding block in such a manner that their identity can be maintained throughout the test.

30.1.2 The overall length of each cartridge is carefully measured and the measurements recorded on the report form. The cartridges are then returned to the recessed holding block to their original position.

30.2 During Firing

30.2.1 One identified and measured cartridge from the test sample shall be place in the cylinder of the test w3eapon. five unmeasured cartridges shall be placed in the cylinder of the weapon. The cartridges are positioned in the cylinder of thee weapon in such a manner that the five unmeasured cartridges will be fired; the measured test cartridge shall not be fired.

30.2.1 One identified and measured cartridge from the test sample shall be placed in the cylinder of the test weapon. Five samples shall be placed in the cylinder of the test weapon. Five unmeasured cartridges shall b placed in the cylinder of the weapon. The cartridges are positioned in the cylinder of the weapon in such a manner that the five unmeasured cartridges will be fired; the measured test cartridges shall not be fired.

30.2.2 The five unmeasured cartridges are then fired form a hand-held revolver and the fired cases removed from the cylinder.

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APPENDIX "I" CONT'D

30.2.3 The identified and measured cartridge shall then be removed from the cylinder and placed in its original position in the recessed holding block.

30.2.4 The procedure prescribed in 30.2.1 through 30.2.3 shall be repeated for each cartridge in the test sample.

30.2.5 After all cartridges in the test sample have been tested, the overall length of each of the test cartridges is carefully measured again and the measurements placed on the report form in such a manner that the measurements can be associated with the initial measurements.

40.0 RECORDING OF DATA

Test sheets shall show the following:

a. Measurements of the overall length of the individual test cartridges, both before and after the test.

b. Elongation, if any has taken place.

40.1 Test weapon data

a. Weapon number.

b. Total number of cartridges fired in weapon.

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CONCLUDING MATERIAL

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1305-AC04)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-46409D (AR)	10 April 1991

3. DOCUMENT TITLE

CARTRIDGE, CALIBER .38 SPECIAL, BALL, M41

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6671 (2) AUTOVON 880-6671
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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